

15.6 Using the NLS panel data on $N = 716$ young women, we consider only years 1987 and 1988. We are interested in the relationship between $\ln(\text{WAGE})$ and experience, its square, and indicator variables for living in the south and union membership. Some estimation results are in Table 15.10.

TABLE 15.10 Estimation Results for Exercise 15.6

	(1) OLS 1987	(2) OLS 1988	(3) FE	(4) FE Robust	(5) RE
<i>C</i>	0.9348 (0.2010)	0.8993 (0.2407)	1.5468 (0.2522)	1.5468 (0.2688)	1.1497 (0.1597)
<i>EXPER</i>	0.1270 (0.0295)	0.1265 (0.0323)	0.0575 (0.0330)	0.0575 (0.0328)	0.0986 (0.0220)
<i>EXPER</i> ²	-0.0033 (0.0011)	-0.0031 (0.0011)	-0.0012 (0.0011)	-0.0012 (0.0011)	-0.0023 (0.0007)
<i>SOUTH</i>	-0.2128 (0.0338)	-0.2384 (0.0344)	-0.3261 (0.1258)	-0.3261 (0.2495)	-0.2326 (0.0317)
<i>UNION</i>	0.1445 (0.0382)	0.1102 (0.0387)	0.0822 (0.0312)	0.0822 (0.0367)	0.1027 (0.0245)
<i>N</i>	716	716	1432	1432	1432

(standard errors in parentheses)

- The OLS estimates of the $\ln(\text{WAGE})$ model for each of the years 1987 and 1988 are reported in columns (1) and (2). How do the results compare? For these individual year estimations, what are you assuming about the regression parameter values across individuals (heterogeneity)?
- The $\ln(\text{WAGE})$ equation specified as a panel data regression model is

$$\ln(\text{WAGE}_{it}) = \beta_1 + \beta_2 \text{EXPER}_{it} + \beta_3 \text{EXPER}_{it}^2 + \beta_4 \text{SOUTH}_{it} + \beta_5 \text{UNION}_{it} + (u_i + e_{it}) \quad (\text{XR15.6})$$

Explain any differences in assumptions between this model and the models in part (a).

- Column (3) contains the estimated fixed effects model specified in part (b). Compare these estimates with the OLS estimates. Which coefficients, apart from the intercepts, show the most difference?
- The F -statistic for the null hypothesis that there are no individual differences, equation (15.20), is 11.68. What are the degrees of freedom of the F -distribution if the null hypothesis (15.19) is true? What is the 1% level of significance critical value for the test? What do you conclude about the null hypothesis?
- Column (4) contains the fixed effects estimates with cluster-robust standard errors. In the context of this sample, explain the different assumptions you are making when you estimate with and without cluster-robust standard errors. Compare the standard errors with those in column (3). Which ones are substantially different? Are the robust ones larger or smaller?
- Column (5) contains the random effects estimates. Which coefficients, apart from the intercepts, show the most difference from the fixed effects estimates? Use the Hausman test statistic (15.36) to test whether there are significant differences between the random effects estimates and the fixed effects estimates in column (3) (Why that one?). Based on the test results, is random effects estimation in this model appropriate?

$$\begin{aligned}
 (a) \quad 1987: & 0.9348 + 0.1270 \text{ EXPER} - 0.0033 \text{ EXPER}^2 - 0.2128 \text{ SOUTH} + 0.1445 \text{ UNION} \\
 1988: & 0.8993 + 0.1265 \text{ EXPER} - 0.0031 \text{ EXPER}^2 - 0.2384 \text{ SOUTH} + 0.1102 \text{ UNION}
 \end{aligned}$$

1988 較負 1988 較小

→ 於單一年份做 OLS 時，假設所有個人的迴歸參數都相同

$$(V_i, \beta_{1i} = \beta_1, \beta_{2i} = \beta_2, \dots, \beta_{5i} = \beta_5)$$

若此時個人存在不可觀見之差異，且和解釋變數有相關，則 OLS 估計可能產生偏誤
 → 個體的固定效果

$$(b) \text{ OLS 中假設 } u_i = 0 \text{ or } \text{Cov}(x_{it}, u_i) = 0 \quad [FE: \text{Cov}(x_{it}, u_i) = 0, RE: \text{Cov}(x_{it}, u_i) \neq 0]$$

⇒ panel model 明確處理了個體異質性，而 OLS 則沒有

(c)

$$FE: \ln(\widehat{WAGE}) = 1.5468 + 0.0575 \text{ EXPER} - 0.0012 \text{ EXPER}^2 - 0.3261 \text{ SOUTH} + 0.0822 \text{ UNION}$$

⇒ EXPER 項的係數差最多 OLS: 0.127, 0.1265, FE: 0.0575

$$(d) H_0: u_1 = u_2 = \dots = u_N, \quad d = N - 1 = 715, \quad df_2 = NT - N - K = 1432 - 716 - 4 = 712$$

$$\Rightarrow F \sim F_{715, 712}, \quad F_{0.99}(715, 712) \approx 1.19 < 11.68$$

∴ we reject $H_0 \Rightarrow$ 個體間存在顯著差異性

(e)

the different assumption is cluster robust allow $\text{Cov}(e_{it}, e_{is}) \neq 0$

⇒ SE 的假設較弱

SE 項差異最大的是 SOUTH，而 robust 的部份大了幾乎 2 倍 $\begin{cases} 0.1258 \\ 0.2495 \end{cases}$

(f)

SOUTH 的差異最大: (FE: -0.3261, RE: -0.2326)

$$\text{Hausman test} = H_0: \text{Cov}(x_{it}, u_i) = 0$$

$$\therefore H = (\hat{\beta}_{FE} - \hat{\beta}_{RE})' [\text{Var}(\hat{\beta}_{FE}) - \text{Var}(\hat{\beta}_{RE})]^{-1} (\hat{\beta}_{FE} - \hat{\beta}_{RE}) = 6.19 \text{ (df=4)}, \text{ and } \chi^2_{0.95, 4} = 9.49$$

∵ $H < 9.49 \Rightarrow$ we do not reject H_0 ∴ RE 的結果是合適的

15.20 This exercise uses data from the STAR experiment introduced to illustrate fixed and random effects for grouped data. In the STAR experiment, children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star*.

- Estimate a regression equation (with no fixed or random effects) where *READSCORE* is related to *SMALL*, *AIDE*, *TCHEXPER*, *BOY*, *WHITE_ASIAN*, and *FREELUNCH*. Discuss the results. Do students perform better in reading when they are in small classes? Does a teacher's aide improve scores? Do the students of more experienced teachers score higher on reading tests? Does the student's sex or race make a difference?
- Reestimate the model in part (a) with school fixed effects. Compare the results with those in part (a). Have any of your conclusions changed? [Hint: specify *SCHID* as the cross-section identifier and *ID* as the "time" identifier.]
- Test for the significance of the school fixed effects. Under what conditions would we expect the inclusion of significant fixed effects to have little influence on the coefficient estimates of the remaining variables?
- Reestimate the model in part (a) with school random effects. Compare the results with those from parts (a) and (b). Are there any variables in the equation that might be correlated with the school effects? Use the LM test for the presence of random effects.
- Using the *t*-test statistic in equation (15.36) and a 5% significance level, test whether there are any significant differences between the fixed effects and random effects estimates of the coefficients on *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, and *FREELUNCH*. What are the implications of the test outcomes? What happens if we apply the test to the fixed and random effects estimates of the coefficient on *BOY*?
- Create school-averages of the variables and carry out the Mundlak test for correlation between them and the unobserved heterogeneity.

a.

```
##
## t test of coefficients:
##
##               Estimate Std. Error   t value   Pr(>|t|)
## (Intercept)  437.76425    1.31238  333.5646 < 2.2e-16 ***
## small        5.82282     0.99601   5.8461 5.308e-09 ***
## aide         0.81784     0.93280   0.8768  0.3807
## tchexper     0.49247     0.07128   6.9090 5.409e-12 ***
## boy         -6.15642     0.79656  -7.7287 1.273e-14 ***
## white_asian  3.90581     0.89098   4.3837 1.187e-05 ***
## freelunch   -14.77134    0.85637 -17.2488 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

aide isn't significant
⇒ it doesn't improve score

tchexper is positive and significant,

⇒ it improve score

sex and race are significant

⇒ both are influence score

b,

```

      Estimate Std. Error    t value    Pr(>|t|)
(Intercept) 411.1626296 2.37842242 172.871996 0.000000e+00
small       6.4902305 0.93102600  6.971052 3.503010e-12
aide       0.9960875 0.86127432  1.156528 2.475141e-01
tchexper    0.2855668 0.07114639  4.013792 6.052000e-05
boy        -5.4559412 0.73104341 -7.463225 9.716472e-14
white_asian 8.0280192 1.53786343  5.220242 1.850422e-07
freelunch  -14.5935724 0.84646565 -17.240596 5.786271e-65

```

white_asian = 3.905 → 8.02

影響更大

其餘的顯著性大致沒有
變化,係數也差不多

c,

```

##
## Model 1: restricted model
## Model 2: readscore ~ small + aide + tchexper + boy + white_asian + freelunch +
##   factor(schid)
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df    F    Pr(>F)
## 1     5759
## 2    5681 78 20.334 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

p-value = 2.2×10^{-16} ⇒ reject H_0 : no fixed effect

學校的固定效果顯著存在

, 不適合用 pooled model

d,

```

##
## t test of coefficients:
##
##      Estimate Std. Error t value Pr(>|t|)
## (Intercept) 436.12677    2.04477 213.2890 < 2.2e-16 ***
## small       6.45872    1.63589   3.9481 7.970e-05 ***
## aide       0.99215    1.38238   0.7177 0.47297
## tchexper    0.30268    0.13570   2.2305 0.02575 *
## boy        -5.51208    0.70447  -7.8244 6.028e-15 ***
## white_asian 7.35048    1.74429   4.2140 2.547e-05 ***
## freelunch  -14.58433    1.05731 -13.7939 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

差異較大
和

⇒ $\text{Cov}(\alpha_i, \alpha_i) \neq 0$ We reject $H_0: \sigma_a^2 = 0$

⇒ the random effect
is exist

e,

```

##
## Hausman Test
##
## data:  readscore ~ small + aide + tchexper + boy + white_asian + freelunch
## chisq = 13.809, df = 6, p-value = 0.03184
## alternative hypothesis: one model is inconsistent

```

reject H_0

⇒ random effect 不一致

```
coef(fe_plm)["boy"]
```

```
##      boy  
## -5.455941
```

```
coef(re_mod)["boy"]
```

```
##      boy  
## -5.512081
```

boy 在 FE 和 RE 中差異不大 $\Rightarrow$ 性別和學校的固定效果
相對較低

FE

RE

```
Estimate  
(Intercept) 411.1626296  
small        6.4902305  
aide         0.9960875  
tchexper     0.2855668  
boy          -5.4559412  
white_asian  8.0280192  
freelunch    -14.5935724
```

```
##      Estimate  
## (Intercept) 436.12677  
## small      6.45872  
## aide       0.99215  
## tchexper   0.30268  
## boy       -5.51208  
## white_asian 7.35048  
## freelunch -14.58433
```

$\rightarrow$ 差異較大

f,

```
##  
## Linear hypothesis test:  
## m_small = 0  
## m_aide = 0  
## m_tchexper = 0  
## m_boy = 0  
## m_white_asian = 0  
## m_freelunch = 0  
##  
## Model 1: restricted model  
## Model 2: readscore ~ small + aide + tchexper + boy + white_asian + freelunch +  
##      m_small + m_aide + m_tchexper + m_boy + m_white_asian + m_freelunch  
##  
## Note: Coefficient covariance matrix supplied.  
##  
##      Res.Df Df      F Pr(>F)  
## 1      5695  
## 2      5689  6 2.2537 0.0356 *  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$\therefore p\text{-value} < 0.05$

we reject $H_0: \text{cov}(x_{ij}, d_i) = 0$

fixed effect is more
suitable than random effect